

MATH 4820 / Regression Analysis

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**Transformations and Weighting
to
Correct Model Inadequacies**



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5.1 INTRODUCTION

- Chapter 4 presented several techniques for checking the adequacy of the linear regression model. Recall that regression model fitting has several implicit assumptions, including the following:
 1. The model errors have mean zero and constant variance and are uncorrelated.
 2. The model errors have a normal distribution — this assumption is made in order to conduct hypothesis tests and construct CIs — under this assumption, the errors are independent.
 3. The form of the model, including the specification of the regressors, is correct.

5.2 VARIANCE STABILIZING TRANSFORMATIONS

Constant variance assumption

- Often violated when the variance is functionally related to the mean.
- Transformation on the response may eliminate the problem.
- The strength of the transformation depends on the amount of curvature that is induced.
- If not satisfied, the regression coefficients will have larger standard errors (less precision)

Relationship of σ^2 to $E(y)$	Transformation
$\sigma^2 \propto \text{constant}$	$y' = y$ (no transformation)
$\sigma^2 \propto E(y)$	$y' = \sqrt{y}$ (square root; Poisson data)
$\sigma^2 \propto E(y)[1 - E(y)]$	$y' = \sin^{-1}(\sqrt{y})$ (arcsin; binomial proportions $0 \leq y_i \leq 1$)
$\sigma^2 \propto [E(y)]^2$	$y' = \ln(y)$ (log)
$\sigma^2 \propto [E(y)]^3$	$y' = y^{-1/2}$ (reciprocal square root)
$\sigma^2 \propto [E(y)]^4$	$y' = y^{-1}$ (reciprocal)

EXAMPLE 5.1: THE ELECTRIC UTILITY DATA

- An electric utility is interested in developing a model relating peak-hour demand (y) to total energy usage during the month (x). This is an important planning problem because while most customers pay directly for energy usage (in kilowatt-hours), the generation system must be large enough to meet the maximum demand imposed.
- Data for 53 residential customers for the month of August are shown in [Table 5.2](#), and a scatter diagram is given in [Figure 5.1](#). As a starting point, a simple linear regression model is assumed, and the least-squares fit is

$$\hat{y} = -0.8313 + 0.00368 x$$

EXAMPLE 5.1: THE ELECTRIC UTILITY DATA

TABLE 5.2 Demand (y) and Energy Usage (x) Data for 53 Residential Customers, August 1979

Customer	x (KWH)	y (KW)	Customer	x (KWH)	y (KW)
1	679	0.79	27	837	4.20
2	292	0.44	28	1748	4.88
3	1012	0.56	29	1381	3.48
4	493	0.79	30	1428	7.58
5	582	2.70	31	1255	2.63
6	1156	3.64	32	1777	4.99
7	997	4.73	33	370	0.59
8	2189	9.50	34	2316	8.19
9	1097	5.34	35	1130	4.79
10	2078	6.85	36	463	0.51
11	1818	5.84	37	770	1.74
12	1700	5.21	38	724	4.10
13	747	3.25	39	808	3.94
14	2030	4.43	40	790	0.96
15	1643	3.16	41	783	3.29
16	414	0.50	42	406	0.44
17	354	0.17	43	1242	3.24
18	1276	1.88	44	658	2.14
19	745	0.77	45	1746	5.71
20	435	1.39	46	468	0.64
21	540	0.56	47	1114	1.90
22	874	1.56	48	413	0.51
23	1543	5.28	49	1787	8.33
24	1029	0.64	50	3560	14.94
25	710	4.00	51	1495	5.11
26	1434	0.31	52	2221	3.85
			53	1526	3.93

EXAMPLE 5.1 CONT...

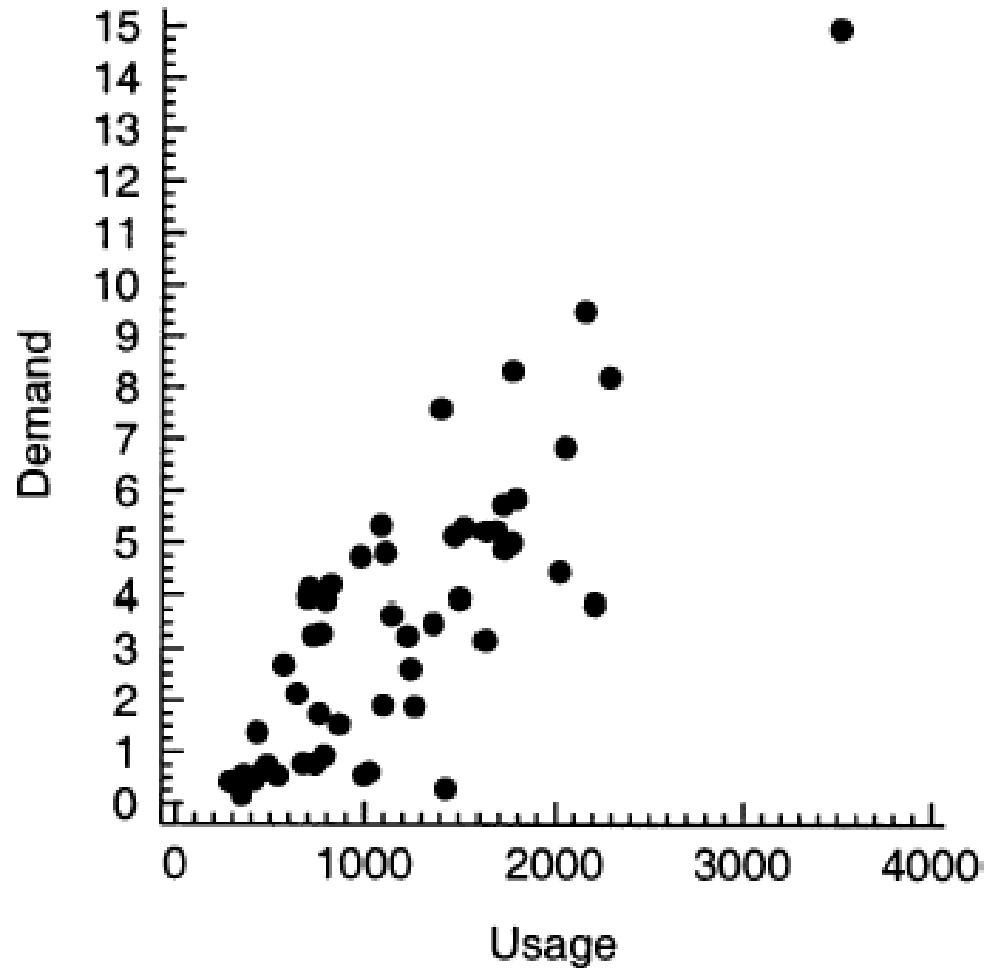


Figure 5.1 Scatter diagram of the energy demand (kW) versus energy usage (kWh), Example 5.1.

EXAMPLE 5.1 CONT...

- Summary of the ANOVA for the regression model

TABLE 5.3 Analysis of Variance for Regression of y on x for Example 5.1

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P Value
Regression	303.6331	1	302.6331	121.66	< 0.0001
Residual	126.8660	51	2.4876		
Total	429.4991	52			

- Summary of the Estimates:

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.8313037  0.4416121  -1.882   0.0655 .
x..kWh.      0.0036828  0.0003339   11.030 4.11e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.577 on 51 degrees of freedom
Multiple R-squared:  0.7046,    Adjusted R-squared:  0.6988
F-statistic: 121.7 on 1 and 51 DF,  p-value: 4.106e-15
```

EXAMPLE 5.1 – SQUARE ROOT TRANSFORMATION TO STABILIZE VARIANCE

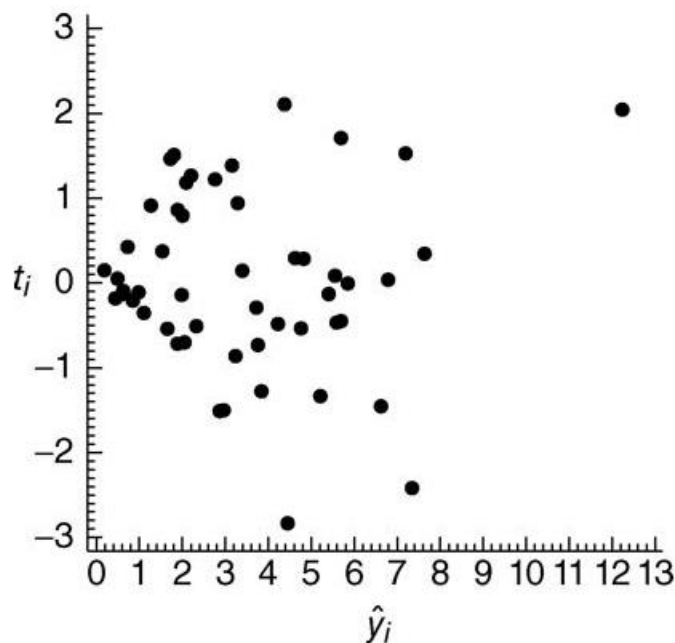


Figure 5.2 Plot of R -student values t_i versus fitted values $\hat{y}_i$, Example 5.1.

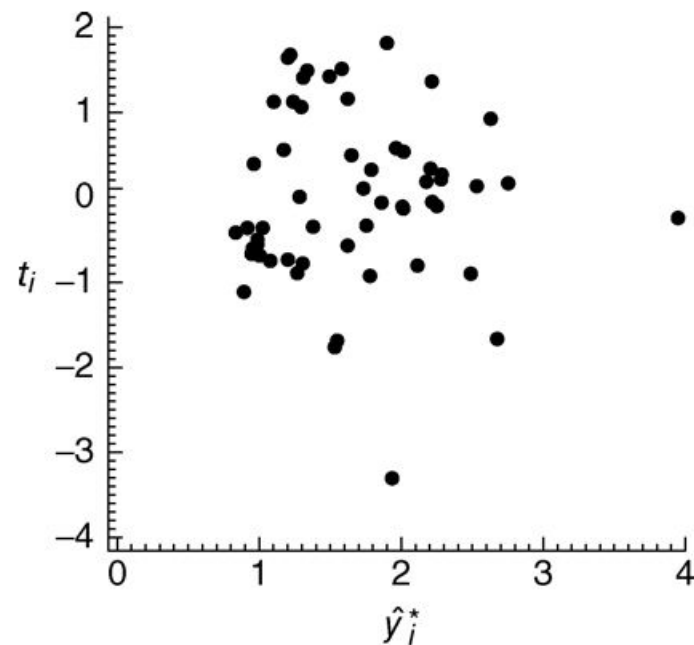


Figure 5.3 Plot of R -student values t_i versus fitted values $\hat{y}_i^*$ for the transformed data,

5.3 TRANSFORMATIONS TO LINEARIZE THE MODEL

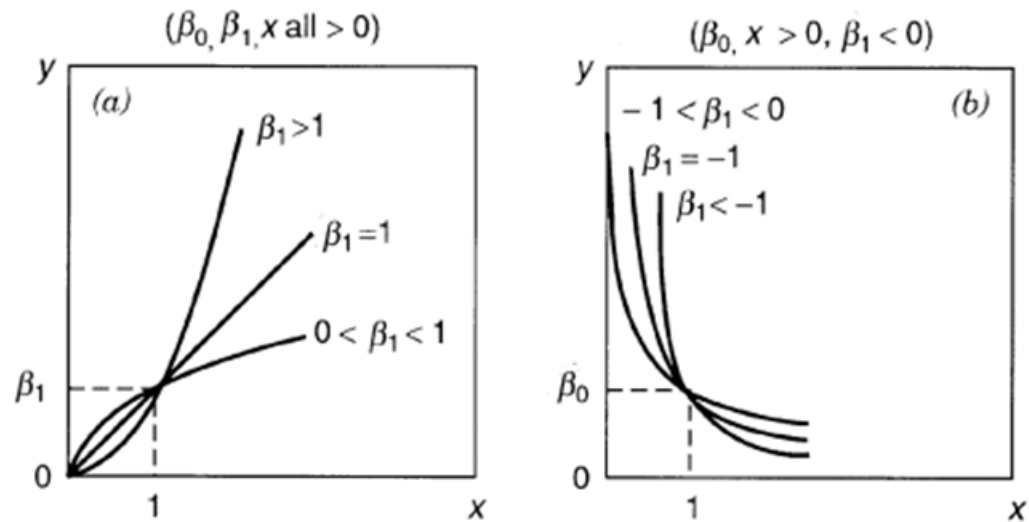
- If a transformation of a nonlinear function can result in a linear function – we say it is *intrinsically* or *transformably* linear.
- Example:

$$y = \beta_0 e^{\beta_1 x} \varepsilon$$

$$\ln y = \ln \beta_0 + \beta_1 x + \ln \varepsilon$$

$$y' = \beta'_0 + \beta_1 x + \varepsilon'$$

$$y = \beta_0 x^{\beta_1}$$



$$y = \beta_0 e^{\beta_1 x}$$

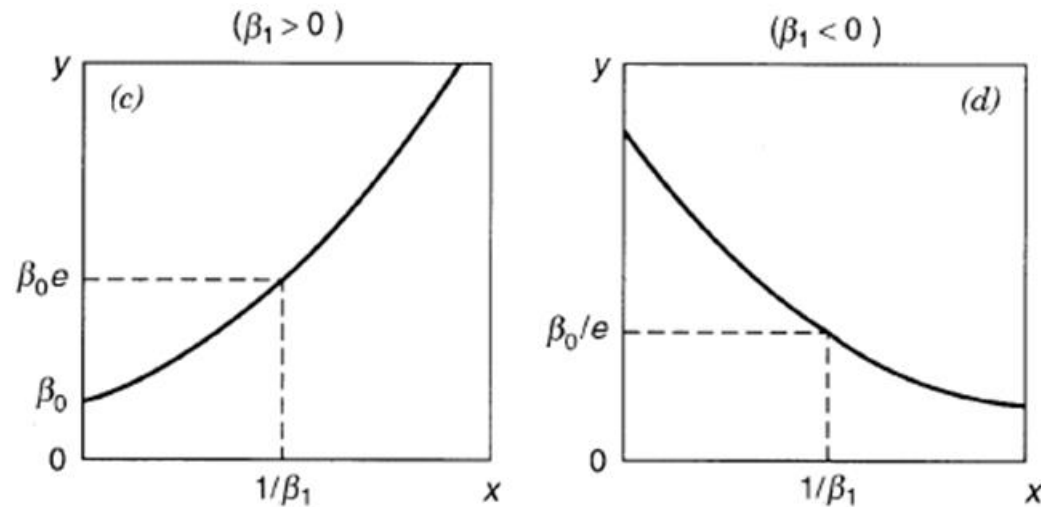
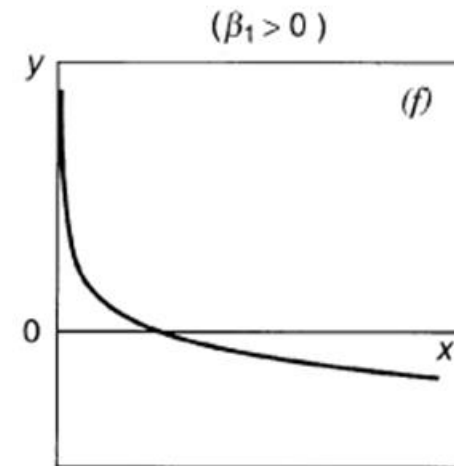
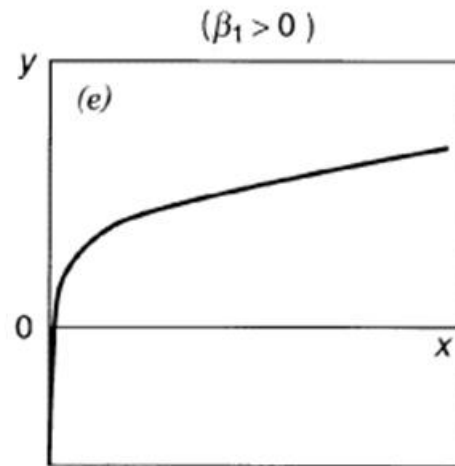


Figure 5.4 (a)-(d). Linearizable Functions [Daniel and Wood (1980)]

$$y = \beta_0 + \beta_1 \log x$$



$$y = \frac{x}{\beta_0 x - \beta_1}$$

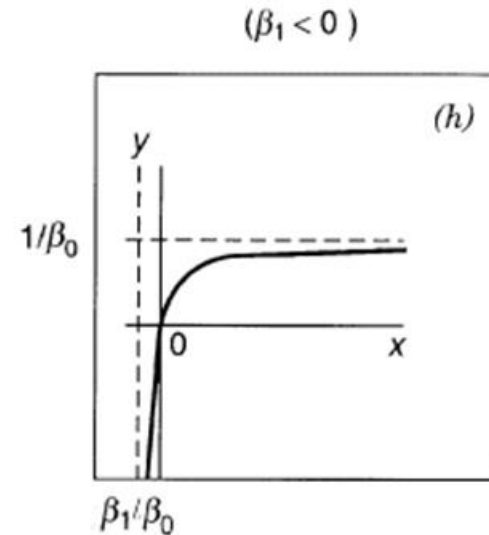
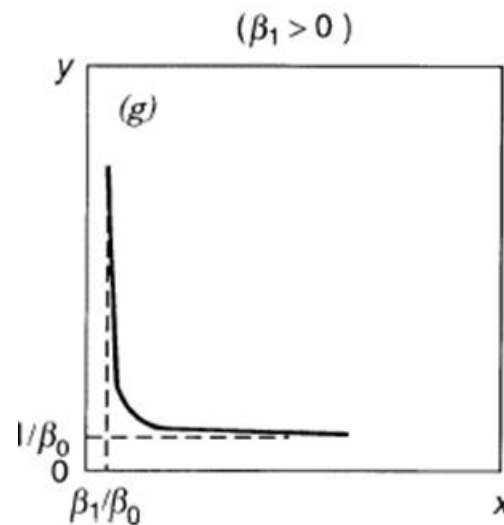


Figure 5.4 (e)-(h). Linearizable Functions [Daniel and Wood (1980)]

5.3 TRANSFORMATIONS TO LINEARIZE THE MODEL

TABLE 5.4 Linearizable Functions and Corresponding Linear Form

Figure	Linearizable Function	Transformation	Linear Form
5.4a, b	$y = \beta_0 x^{\beta_1}$	$y' = \log y, x' = \log x$	$y' = \log \beta_0 + \beta_1 x'$
5.4c, d	$y = \beta_0 e^{\beta_1 x}$	$y' = \ln y,$	$y' = \ln \beta_0 + \beta_1 x$
5.4e, f	$y = \beta_0 + \beta_1 \log x$	$x' = \log x$	$y' = \beta_0 + \beta_1 x'$
5.4g, h	$y = \frac{x}{\beta_0 x - \beta_1}$	$y' = \frac{1}{y}, x' = \frac{1}{x}$	$y' = \beta_0 - \beta_1 x'$

EXAMPLE 5.2: THE WINDMILL DATA

- A research engineer is investigating the use of a windmill to generate electricity. He has collected data on the DC output from his windmill and the corresponding wind velocity

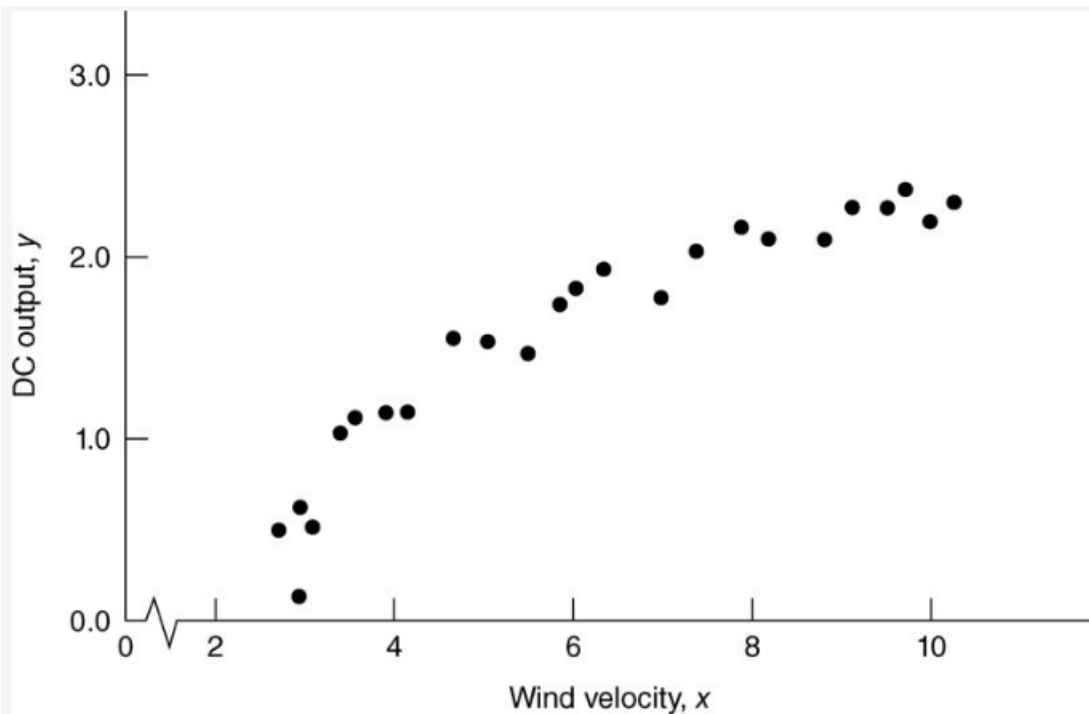


Figure 5.5 Plot of DC output y versus wind velocity x for the windmill data.

Observation Number, i	Wind Velocity, x_i (mph)	DC Output, y_i
1	5.00	1.582
2	6.00	1.822
3	3.40	1.057
4	2.70	0.500
5	10.00	2.236
6	9.70	2.386
7	9.55	2.294
8	3.05	0.558
9	8.15	2.166
10	6.20	1.866
11	2.90	0.653
12	6.35	1.930
13	4.60	1.562
14	5.80	1.737
15	7.40	2.088
16	3.60	1.137
17	7.85	2.179
18	8.80	2.112
19	7.00	1.800
20	5.45	1.501
21	9.10	2.303
22	10.20	2.310
23	4.10	1.194
24	3.95	1.144
25	2.45	0.123

EXAMPLE 5.2 THE WINDMILL DATA

- A straight line fit to the data resulted in:

$$\hat{y} = 0.1309 + 0.2411x$$

- The summary statistics are

$$R^2 = 0.8745,$$

$$MS_{\text{Res}} = 0.0557, \text{ and}$$

$$F_0 = 160.26 \text{ (with a P-value} < 0.0001\text{)}$$

Analysis of Variance Table

Response: DC.Output..yi

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Wind.Velocity..xi..mph.	1	8.9296	8.9296	160.26	7.546e-12 ***
Residuals	23	1.2816	0.0557		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

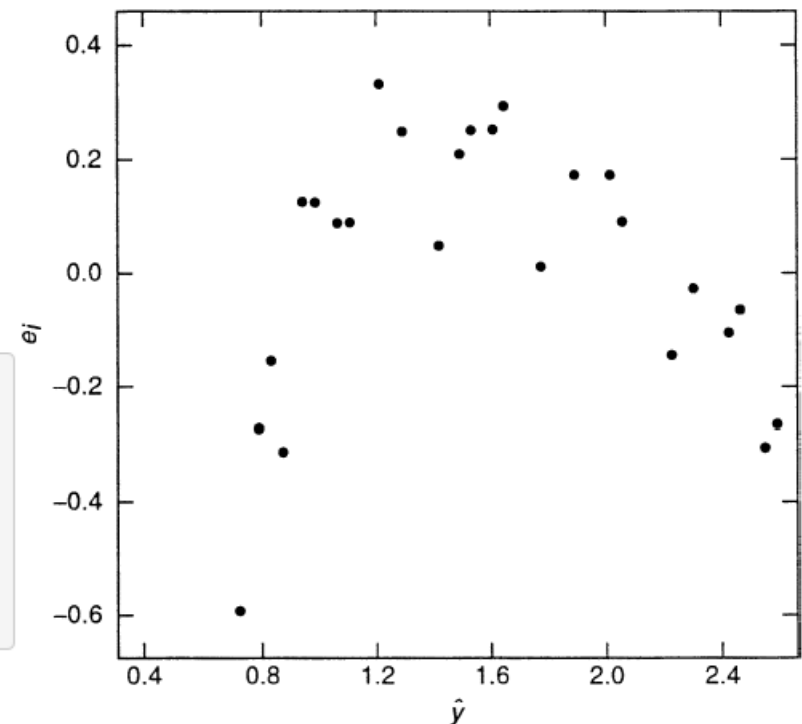


Figure 5.6 Plot of residuals e_i versus fitted values $\hat{y}_i$ for the windmill data.

EXAMPLE 5.2 THE WINDMILL DATA

- We might initially consider using a quadratic model
 - [Figure 5.5](#) suggests that as wind speed increases, DC output approaches an upper limit of approximately 2.5
 - This is also consistent with the theory of windmill operation.
 - Since the quadratic model will eventually bend downward as wind speed increases, it would not be appropriate for these data.

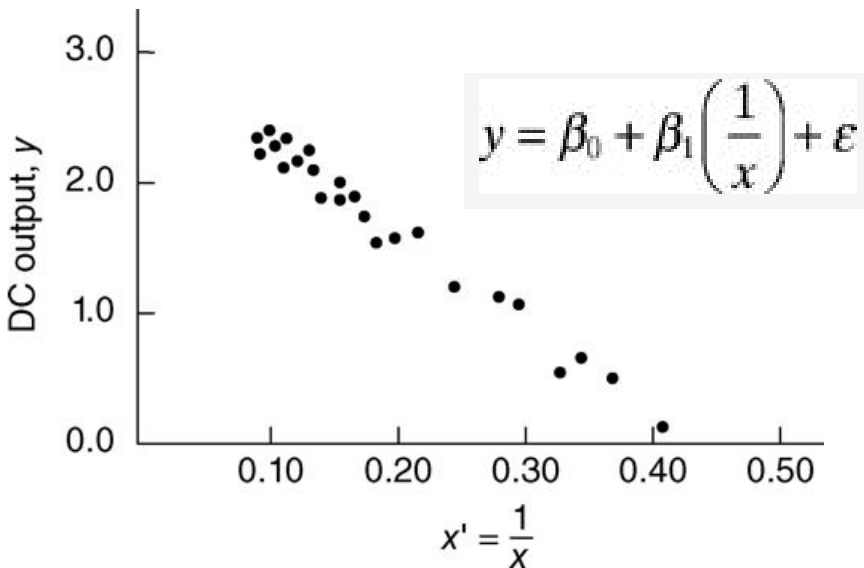


Figure 5.7 Plot of DC output versus $x' = 1/x$ for the windmill data.

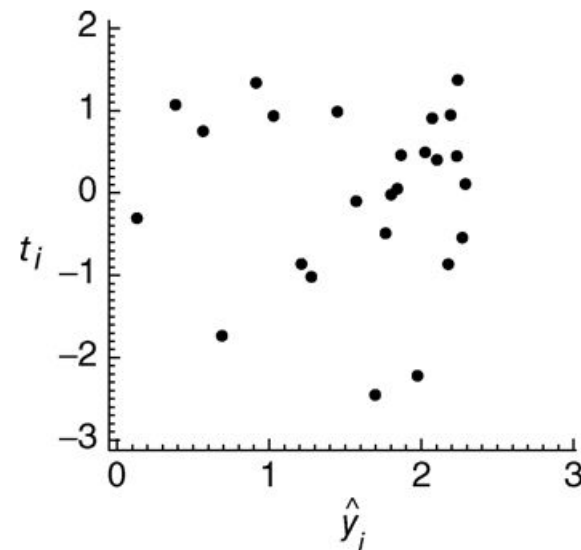


Figure 5.8 Plot of R -student values t_i versus fitted values $\hat{y}_i$ for the transformed model for the windmill data.

EXAMPLE 5.2 THE WINDMILL DATA

Using a reciprocal transformation, the resulting fitted model:

$$\hat{y} = 2.9789 - 6.9345x'$$

The summary statistics are

$$R^2 = 0.9800,$$

$$MS_{\text{Res}} = 0.0089, \text{ and}$$

$$F_0 = 1128.43 \text{ (with a P-value} < 0.0001\text{)}$$

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.9789      0.0449   66.34  <2e-16 ***
x.inv         -6.9345      0.2064  -33.59  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09417 on 23 degrees of freedom
Multiple R-squared:  0.98,    Adjusted R-squared:  0.9792
F-statistic: 1128 on 1 and 23 DF,  p-value: < 2.2e-16
```

```
Analysis of Variance Table

Response: DC.Output..yi
              Df Sum Sq Mean Sq F value    Pr(>F)
x.inv          1 10.007 10.0072 1128.4 < 2.2e-16 ***
Residuals     23  0.204  0.0089
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5.3 TRANSFORMATIONS TO LINEARIZE THE MODEL

Note:

- Least-squares estimator has least-squares properties with respect to the transformed data, not original data.

5.4 ANALYTICAL METHODS FOR SELECTING A TRANSFORMATION

5.4.1 TRANSFORMATIONS ON Y: THE BOX-COX METHOD

- Suppose that we wish to transform y to correct nonnormality and/or nonconstant variance.
- A useful class of transformations is the **power transformation**, y^λ where λ is a parameter to be determined.
- The parameters of the regression model and λ can be estimated simultaneously using the method of maximum likelihood.

5.4.1 TRANSFORMATIONS ON Y:

THE BOX-COX METHOD

- The appropriate procedure to be used is

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda \dot{y}^{\lambda-1}}, & \lambda \neq 0 \\ \dot{y} \ln y, & \lambda = 0 \end{cases}$$

where $\dot{y} = \ln^{-1}[(1/n \sum_{i=1}^n \ln y_i)]$ is the **geometric mean** of the observations.

- The model to be fit is

$$y^{(\lambda)} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

EXAMPLE 5.3 THE ELECTRIC UTILITY DATA

TABLE 5.7 Values of the Residual Sum of Squares for Various Values of λ , Example 5.3

λ	$SS_{\text{Res}}(\lambda)$
-2	34,101.0381
-1	986.0423
-0.5	291.5834
0	134.0940
0.125	118.1982
0.25	107.2057
0.375	100.2561
0.5	96.9495
0.625	97.2889
0.75	101.6869
1	126.8660
2	1275.5555

EXAMPLE 5.3 THE ELECTRIC UTILITY DATA

If we take $\lambda = 0.5$ as the optimum value, then an **approximate 95% confidence interval for λ** may be found by calculating the critical sum of squares SS^* from Eq. 5.5 as follows:

$$\begin{aligned} SS^* &= SS_{\text{Res}}(\hat{\lambda}) e^{\chi_{0.05, 1}^2 / n} \\ &= 96.9495 e^{3.84/53} \\ &= 96.9495 (1.0751) \\ &= 104.23 \end{aligned}$$

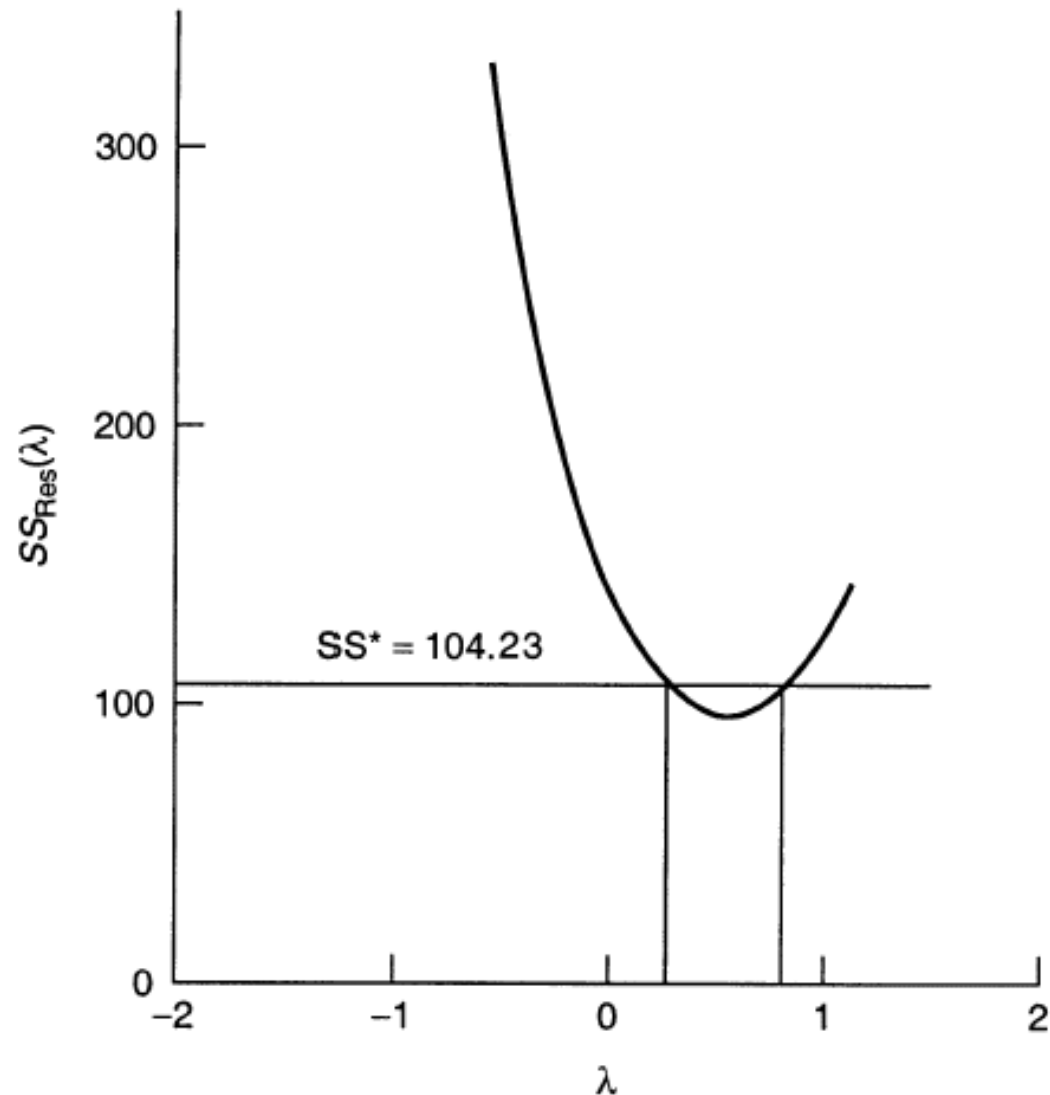


Figure 5.10 Plot of residual sum of squares $SS_{\text{Res}}(\lambda)$ versus λ .

5.5 GENERALIZED AND WEIGHTED LEAST SQUARES

- Linear regression models with nonconstant error variance can also be fitted by the method of **weighted least squares**.
- In this method of estimation the deviation between the observed and expected values of y_i is multiplied by a **weight** w_i chosen inversely proportional to the variance of y_i .

5.5 GENERALIZED AND WEIGHTED LEAST SQUARES

- In simple linear regression, the weighted least squares function is

$$S(\beta_0, \beta_1) = \sum_{i=1}^n w_i (y_i - \beta_0 - \beta_1 x_i)^2$$

- The resulting least squares normal equations:

$$\hat{\beta}_0 \sum_{i=1}^n w_i + \hat{\beta}_1 \sum_{i=1}^n w_i x_i = \sum_{i=1}^n w_i y_i$$

$$\beta_0 \sum_{i=1}^n w_i x_i + \hat{\beta}_1 \sum_{i=1}^n w_i x_i^2 = \sum_{i=1}^n w_i x_i y_i$$

5.5.1 GENERALIZED LEAST SQUARES

- When the model under investigation is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$E(\boldsymbol{\varepsilon}) = \mathbf{0}, \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$$

then ordinary least squares estimation is no longer appropriate.

- We will approach this problem by transforming the model to a **new set** of observations that satisfy the standard least-squares assumptions.
- Ordinary least squares can then be applied to the new set of data.

5.5.1 GENERALIZED LEAST SQUARES

TABLE 5.8 Analysis of Variance for Generalized Least Squares

Source	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Regression	$SS_R = \hat{\beta}' \mathbf{B}' \mathbf{z}$ $= \mathbf{y}' \mathbf{V}^{-1} \mathbf{X} (\mathbf{X}' \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}' \mathbf{V}^{-1} \mathbf{y}$	p	SS_R/p	MS_R/MS_{Res}
Error	$SS_{Res} = \mathbf{z}' \mathbf{z} - \hat{\beta}' \mathbf{B}' \mathbf{z}$ $= \mathbf{y}' \mathbf{V}^{-1} \mathbf{y}$ $\quad - \mathbf{y}' \mathbf{V}^{-1} \mathbf{X} (\mathbf{X}' \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}' \mathbf{V}^{-1} \mathbf{y}$	$n - p$	$SS_{Res}/(n - p)$	
Total	$\mathbf{z}' \mathbf{z} = \mathbf{y}' \mathbf{V}^{-1} \mathbf{y}$	n		

5.5.1 GENERALIZED LEAST SQUARES

- Let $\sigma^2\mathbf{V}$ represent the variance-covariance matrix of the errors.
- The least squares normal equations are

$$(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{V}^{-1}\mathbf{y}$$

- The solution is then

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y}$$

5.5.2 WEIGHTED LEAST SQUARES

When the errors ε are uncorrelated but have unequal variances so that the covariance matrix of ε is

$$\sigma^2 \mathbf{V} = \sigma^2 \begin{bmatrix} \frac{1}{w_1} & & & 0 \\ & \frac{1}{w_2} & & \\ & & \ddots & \\ 0 & & & \frac{1}{w_n} \end{bmatrix}$$

say, the estimation procedure is usually called **weighted least squares**.

- The weighted least squares equations are

$$(\mathbf{X}'\mathbf{W}\mathbf{X}) \hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{W}\mathbf{y}$$

- The **weighted least-squares estimator** is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1} \mathbf{X}'\mathbf{W}\mathbf{y}$$

5.5.2 WEIGHTED LEAST SQUARES

USING LS FORMULAR

$$\hat{\beta} = (X'WX)^{-1}X'Wy$$

$$\mathbf{B} = \begin{bmatrix} 1\sqrt{w_1} & x_{11}\sqrt{w_1} & \cdots & x_{1k}\sqrt{w_1} \\ 1\sqrt{w_2} & x_{21}\sqrt{w_2} & \cdots & x_{2k}\sqrt{w_2} \\ \vdots & \vdots & & \vdots \\ 1\sqrt{w_n} & x_{n1}\sqrt{w_n} & \cdots & x_{nk}\sqrt{w_n} \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} y_1\sqrt{w_1} \\ y_2\sqrt{w_2} \\ \vdots \\ y_n\sqrt{w_n} \end{bmatrix}$$

- The **weighted least-squares estimator** is

$$\hat{\beta} = (\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{z} = (X'WX)^{-1}X'Wy$$

5.5.3 SOME PRACTICAL ISSUES

- The weights, w_i , must be known:
 - Prior knowledge or information
 - Residual analysis
 - Chosen inversely proportional to variances of measurement error
 - Guess at the weights and reiteratively estimate them.
- What if weighted least squares is not implemented, but should have been?
 - If ordinary least squares is used in this case, the resulting estimator is still unbiased
 - However, the ordinary least-squares estimator is no longer a minimum-variance estimator.

Example 5.5 Weighted Least Squares

The average monthly income from food sales and the corresponding annual advertising expenses for 30 restaurants are shown in columns a and b of [Table 5.9](#). Management is interested in the relationship between these variables, and so a linear regression model relating food sales y to advertising expense x is fit by ordinary least squares, resulting in $\hat{y} = 49,443.3838 + 8.0484x$. The residuals from this least-squares fit are plotted against $\hat{y}_i$ in [Figure 5.10](#). This plot indicates violation of the constant-variance assumption. Consequently, the ordinary least-squares fit is inappropriate.

Obs. i	(a) Income, Y_i	(b) Advertising Expense, X_i	(c) $\bar{x}$	(d) s_y^2	(e) Weights, w_i
1	81,464	3,000	3,078.3	26,794,616	6.21771 E-08
2	72,661	3,150			5.79507 E-08
3	72,344	3,085			5.97094 E-08
4	90,743	5,225	5,287.5	30,772,013	2.98667 E-08
5	98,588	5,350			2.90195 E-08
6	96,507	6,090			2.48471 E-08
7	126,574	8,925	8,955.0	52,803,695	1.60217 E-08
8	114,133	9,015			1.58431 E-08
9	115,814	8,885			1.61024 E-08
10	123,181	8,950	8,950.0	52,803,695	1.59717 E-08
11	131,434	9,000			1.58726 E-08
12	140,564	1,1345	12,171.0	59,646,475	1.22942 E-08
13	151,352	12,275			1.12852 E-08
14	146,926	12,400			1.11621 E-08
15	130,963	12,525	12,525.0	59,646,475	1.10416 E-08
16	144,630	12,310			1.12505 E-08
17	147,041	13,700			1.00246 E-08
18	179,021	15,000	15,095.0	120,571,061	9.09750 E-09
19	166,200	15,175			8.98563 E-09
20	180,732	14,995			9.10073 E-09
21	178,187	15,050	15,050.0	120,571,061	9.06525 E-09
22	185,304	15,200			8.96987 E-09
23	155,931	15,150			9.00144 E-09
24	172,579	16,800	16,650.0	132,388,992	8.06478 E-09
25	188,851	16,500			8.22030 E-09
26	192,424	17,830			7.57287 E-09
27	203,112	19,500	19,262.5	138,856,871	6.89136 E-09
28	192,482	19,200			7.00460 E-09
29	218,715	19,000			7.08218 E-09
30	214,317	19,350			6.94752 E-09

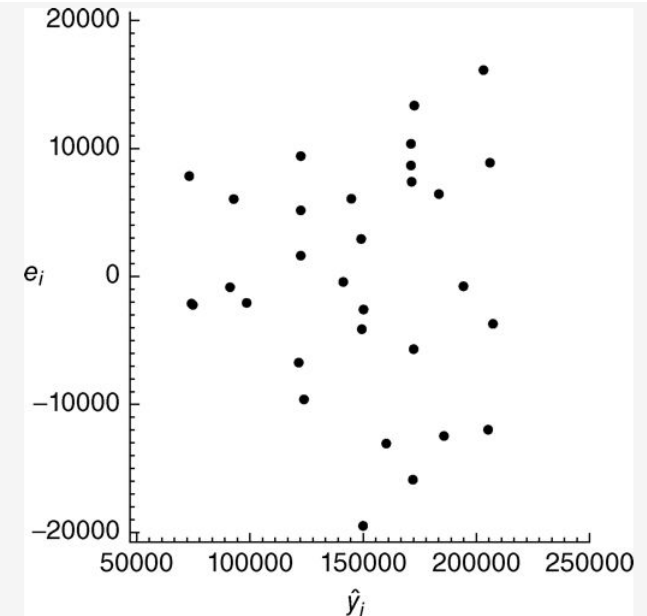


Figure 5.10 Plot of ordinary least-squares residuals versus fitted values, [Example 5.5](#).

$$\hat{s}_y^2 = -9,226,002 + 7781.626\bar{x}$$

EXAMPLE 5.5 WEIGHTED LEAST SQUARES

- Applying weighted least squares to the data, using the weights in Table 5.9 gives the fitted model

$$\hat{y} = 50,975.5667 + 7.9222x$$

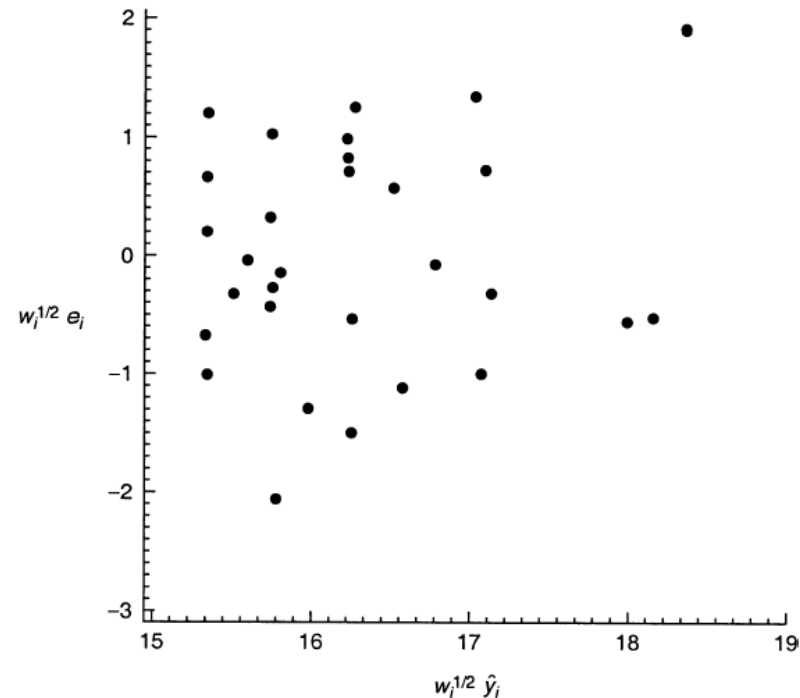


Figure 5.12 Plot of weighted residuals $w_i^{1/2} e_i$ versus weighted fitted values $w_i^{1/2} \hat{y}_i$, Example 5.5.

QUESTIONS?

- ANY QUESTION?

RANDOM EFFECTS

- Regression models can have random effects (that is, factor levels selected at random), and they represent another source of variability
- REML (residual maximum likelihood) is the preferred method of analysis
- REML will estimate the variance component associated with the random effect
- REML is a two-stage iterative procedure
- Implemented in JMP and SAS

Example 5.7 The Delivery Time Data Revisited

We introduced the delivery time data in Example 3.1. In Section 4.2.6 we observed that the first seven observations were collected from San Diego, observations 8–17 from Boston, observations 18–23 from Austin, and observations 24 and 25 from Minneapolis.

It is not unreasonable to assume that the cities used in this study represent a random sample of cities across the country. Ultimately, our interest is the impact of the number of cases delivered and the distance required to make the delivery on the delivery times over the entire country. As a result, a proper analysis needs to consider the impact of the random city effect on this analysis.

Figure 5.12 summarizes the analysis from JMP. We see few differences in the parameter estimates between the mixed model analysis that did not include the city's factor, given in Example 3.1. The P values for cases and distance are larger but only slightly so. The intercept P value is quite a bit larger. Part of this change is due to the significant decrease in the effective degrees of freedom for the intercept effect as the result of using the city information. The plot of the actual delivery times versus the predicted shows that the model is reasonable. The variance component for city is approximately 2.59. The variance for the residual error is 8.79. In the original analysis of Example 3.1 is 10.6. Clearly, part of the variability from the Example 3.1 analysis considered purely random is due to systematic variability due to the various cities, which the REML reflects through the cities' variance component.

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob> t
Intercept	2.0754319	1.356914	6.667	1.53	0.1721
cases	1.7148234	0.1868	21.97	9.18	<.0001*
dist	0.0120317	0.003797	21.9	3.17	0.0045*

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
city	0.2946232	2.5897428	3.4964817	-4.263235	9.442721	22.757
Residual		8.7900161	2.8169566	5.1131327	18.546137	77.243
Total		11.379759				100.000

-2 LogLikelihood = 136.68398351

Figure 5.12 JMP results for the delivery time data treating city as a random effects.